

Pinny

User guide: reviewing drawings and training the models

For reviewers and admins | October 2026 | Pinny training-site 77c9b5e

This guide has two parts. **Part 1** is for everyone who reviews drawings: signing in, scanning, and marking each pin Correct, Wrong or Missed. **Part 2** is for admins: members, the dashboard, building datasets, training the two models, and promoting them.

Site address: <https://pinny.ecinc.us> (production) and <https://staging.pinny.ecinc.us> (staging, for trying changes first).

Hours: the server runs 7 AM to 7 PM Eastern every day and is switched off at night.

Screenshots come from the current version of Pinny running a small test drawing, so your drawings and counts will look different.

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Part 1 - Reviewing drawings

1. What Pinny does

Pinny finds electrical devices (receptacles and other legend symbols) on drawing PDFs and puts a **pin** on each one it finds. You check every pin. Each check does two jobs: it makes that drawing's count right, and it becomes a training example that teaches Pinny's models to find devices better.

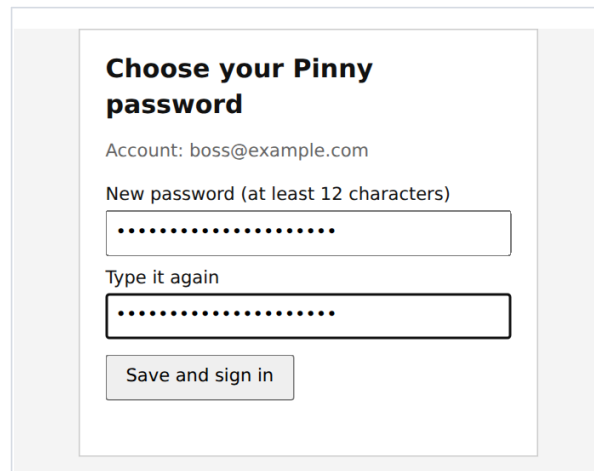
Pinny finds devices three ways today, with no training needed:

- **Legend and whole-set scan** - Pinny reads the drawing set's symbol legend, then looks for every legend symbol on every sheet and counts them by type. Best for complete drawing sets.
- **Template matching** - you draw a box around one example symbol and Pinny finds look-alikes on the page, at any of the four rotations, mirrored, and somewhat smaller or larger.
- **Vector matching** - for PDFs exported from CAD, Pinny compares the actual line drawing rather than a picture, which is more exact.

Two learned models come later, once your team has reviewed enough pins (Part 2): the **verifier**, which double-checks every template match and throws out false ones, and the **point detector**, which finds devices on its own with no template.

2. Signing in

- 1 Your admin adds you and sends you a **one-time set-password link** (valid for 72 hours).
- 2 Open the link, choose a password of at least 12 characters (a password manager helps) and type it twice.
- 3 You are signed in. From then on, go to the site address and sign in with your email and password.

A screenshot of a web form titled "Choose your Pinny password". The form is centered on a light gray background. It contains the following elements: the title "Choose your Pinny password" in bold; the text "Account: boss@example.com"; the label "New password (at least 12 characters)" above a password input field with masked dots; the label "Type it again" above a second password input field with masked dots; and a "Save and sign in" button at the bottom.

The set-password page that the one-time link opens.

Tip. Forgot your password? Ask an admin for a new link (Reset password in the Members panel). A wrong password too many times blocks sign-in for a few minutes.

You stay signed in for up to a week of inactivity (30 days at most). Use **Sign out** at the top of the panel on a shared computer.

3. Open a drawing

- 1 In the **Viewer**, under **1. Document**, press **Choose File** and pick a PDF (up to 200 MB), or choose a drawing someone already uploaded from the **open** list.
- 2 Wait for "name.pdf: N page(s)." The pages appear under **2. Page**. Everyone on the team sees every uploaded drawing. Only the person who uploaded a drawing, or an admin, can press **Delete this drawing**.

4. Scan for devices

Pick one of the two ways below. For a full set with a symbol legend, use A. For a single sheet, or when the legend can't be read, use B.

A. Whole set, using the legend (recommended for drawing sets)

- 1 Under **Legend: count the whole set**, press **Find legend**. Pinny finds the legend page and reads its rows (symbol, tag, description). If it picked the wrong page, open the legend page and press **Read legend on this page**.
- 2 Open **Legend symbols** and check each row. Click a row to fix its name or tag, press **Looks right**, **Split** a row that holds two symbols, or **Add a missing symbol** by drawing a box around it.
- 3 Press **Confirm legend**.
- 4 Under **Whole set**, press **Scan whole set**. Pinny scans every sheet and lists each one with how many symbols it found and how many are left to review.
- 5 Press **Next sheet to review** and review that sheet's pins (sections 5-7). Repeat until every sheet is marked reviewed.

1. Document

Choose File
No file chosen
or

open
set.pdf (4 p, 2112cc33)

set.pdf: 4 page(s).

Delete this drawing

Legend: count the whole set

Pinny reads the drawing's symbol legend, then finds every symbol on every sheet.

Find legend again
Read legend on page 1

Legend on page 1: 5 symbol(s), 4 counted. Confirmed.

► Legend symbols

Legend confirmed

► Engineer's standard legend

Whole set

Scan whole set again

Scanned 3 sheet(s) and found 24 symbol(s) in 1 s. Review each sheet, then mark it fully reviewed.

Sheet	Found	To review	Status
2	8	0	ready to mark reviewed
3	8	0	ready to mark reviewed
4	8	8	to review

Next sheet to review

2. Page

Page 1
Page 2
Page 3
Page 4

ELECTRICAL SYMBOLS LEGEND

SYMBOL	TAG	DESCRIPTION
	D	DUPLEX RECEPTACLE
	G	GFCI RECEPTACLE
	DD	DEDICATED RECEPTACLE
	Q	QUAD RECEPTACLE
	T	THERMOSTAT (BY MECH)

GENERAL NOTES

1. ALL RECEPTACLES TAMPER RESISTANT.

A confirmed legend with five symbols, and the whole-set results: two sheets ready to mark reviewed, one still to review.

B. One symbol, using a template box

- 1 Pick the page under **2. Page** and zoom in (mouse wheel, + / -, or pinch on a tablet) until one receptacle is clear.
- 2 Choose **Template box (T)** in the toolbar and drag a snug rectangle around that one symbol.
- 3 Leave **Threshold** at 0.65 to start. Lower finds more but adds false hits; higher finds fewer.
- 4 Press **Scan this page**, or type pages in **Pages to scan** (blank = all) and press **Scan all pages** to scan the set with the same template. Batch progress shows under **Batch**; press **Next mark to review** to step through every page's pins.

Pinny

Viewer [Dashboard](#) [Label queue](#) [Models](#)

Signed in as boss@example.com (admin) · [Sign out](#)

1. Document

No file chosen
 or
 open

floor-plan.pdf: 1 page(s).

Legend: count the whole set

Pinny reads the drawing's symbol legend, then finds every symbol on every sheet.

2. Page

▼ 3. Scan with a template

Scan mode

Template matching only (no learned model).

Choose **Template box** mode and drag a rectangle around one receptacle.

Template: x 278, y 278, 44 x 44 px

Threshold

☐ Pan / pick pin (V)
 ☒ Template box (T)
 ☐ Mark missed (P)
 200%



Template box mode: a box drawn around one receptacle ("Template: x 278, y 278, 44 x 44 px" in the panel).

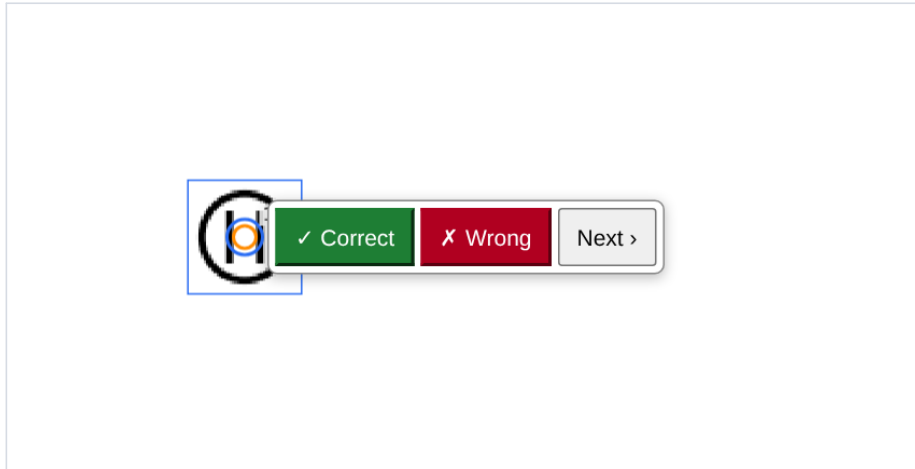
Note. Scanning again creates a new scan with fresh, unreviewed pins. The earlier scan and its reviews are kept and can be reopened from the Scan list.

5. Review each pin: Correct or Wrong

Pins are drawn on the page in colour:

Colour	Meaning
Orange	Not reviewed yet
Green	Approved (Correct)
Magenta	Added by hand (a missed device)
Grey x	Rejected (Wrong) - hidden unless "Show rejected and removed pins" is ticked
Dashed ring / red ring	Still saving / did not save (see section 9)

- 1 Click a pin (in **Pan / pick pin (V)** mode). A small popup appears beside it.
- 2 Press **Correct** (key **A**) if it really is that device, or **Wrong** (key **X**) if not.
- 3 Press **Next** (key **N**) to jump to the next unreviewed pin. The view follows it.



The popup beside a selected pin.

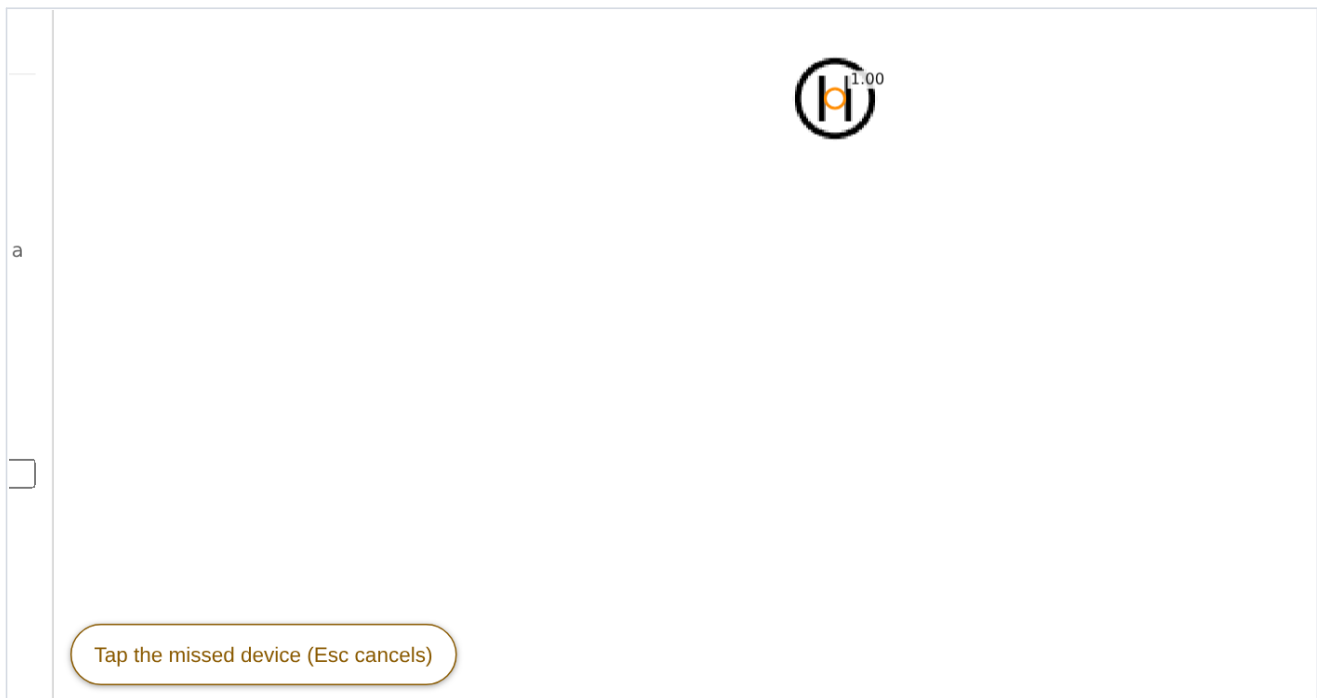
Made a mistake? The pin stays selected after you press a button, so you can press the other one right away. For a pin you added by hand the popup offers **Delete** instead of Wrong.

Tip. Be strict and consistent. A pin is Correct only if it sits on a real device of the type being counted. Text, tags, look-alike symbols and crossed-out devices are Wrong. These clicks are exactly what the models learn from.

6. Mark missed devices

A device the scan did not find has no pin, so marking it has its own button on the drawing.

- 1 Press + **Missed** (bottom-left of the drawing), or the key **M**.
- 2 Click (or tap) the centre of the device the scan missed. A magenta pin appears and Pinny goes back to reviewing with the new pin selected.
- 3 Repeat for each missed device. To cancel, press + Missed again or **Esc**.



The + Missed button while marking: tap the missed device, or press Esc to cancel.

Marking many in a row? Choose **Mark missed (P)** in the toolbar: it stays on for every click until you switch back to Pan (V).

7. Mark the page fully reviewed

When every device on a page has a correct pin and no orange (unreviewed) pins are left, press **Mark page fully reviewed** in the side panel (or on the Label queue).

Important. Only fully reviewed pages can train the point detector, so this step matters. Mark a page only when you have also added every missed device. If someone rescans the page later, it reopens for review.

8. The Label queue

Label queue (top of the panel, or on the training pages) lists pages with the **most uncertain pins first**: the ones whose scores sit closest to the threshold. Reviewing these teaches the models the most per click.

- 1 Open **Label queue**.
- 2 Press **Review** on a page. The viewer opens on its most uncertain pin.
- 3 Review the page, then come back and press **Mark fully reviewed** when its status says "ready to mark".

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Label queue

Pages with the most uncertain pins first. Review opens the viewer on that pin; mark a page fully reviewed once every receptacle on it is pinned and no pin is left unreviewed.

Drawing	Page	Unreviewed	Most uncertain	Status
floor-plan.pdf	1	3	0.200	needs review Review Mark fully reviewed

Most uncertain pins

Drawing	Page	Score	Distance from threshold
floor-plan.pdf	1	1.000	0.200
floor-plan.pdf	1	1.000	0.200
floor-plan.pdf	1	1.000	0.200

The Label queue: pages needing review, and the most uncertain pins.

9. Saving, shortcuts and tips

Every click is saved automatically. The panel shows "Saving N edits...", "All changes saved" or "N edits not saved". Edits survive a page reload. If something doesn't save, press **Retry** (or **Discard**) next to it. If your sign-in expires, Pinny asks you to sign in again and then sends the waiting edits.

Key	Action	Key	Action
A	Correct (approve)	V	Pan / pick pin
X or Delete	Wrong (reject) / delete a hand-added pin	T	Template box
M	Mark one missed device	P	Mark missed (stays on)
N	Next unreviewed pin	B	Next mark in a batch
+ / - / 0	Zoom in / out / fit	R	Rotate the view 90 degrees
H	Hide pins	Esc	Cancel / deselect

- Hold the space bar and drag, or drag with the middle mouse button, to pan in any mode.
- On a tablet: drag with one finger to pan, pinch to zoom, double-tap to zoom in.
- Finish your session before 7 PM Eastern: the server switches off then.

Part 2 - Admins: training the models

Admins see extra pages along the top of the training site: **Datasets**, **Training runs** and **Team**, plus the **Members** and **OCR** panels at the bottom of the viewer.

10. Add and manage members

- 1 In the viewer, scroll to **Members (admin)**.
 - 2 Type the person's email, choose **Reviewer** or **Admin**, and press **Add or change**.
 - 3 Copy the one-time set-password link that appears and send it to them privately (text or in person). It is valid for 72 hours.
 - 4 For a forgotten password, press **Reset password** on their row and send the new link. To take someone off Pinny, press **Remove**. Both end that person's sessions at once.
- Reviewers can upload, scan and review. Admins also manage members, delete anyone's uploads, and train and promote models. Keep admins to one or two people.

11. Watch progress: Dashboard and Team

The **Dashboard** counts labels and pages and shows a **Readiness** card for each model.

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Signed in as bos

Dashboard

2

positive labels

0

negative labels

3

pins not reviewed yet

2 / 5

pins reviewed

0 / 1

pages marked fully reviewed

0

pages usable for the detector

1

drawings uploaded

1

drawings with labels

Readiness

Guidance before training; an admin may train earlier. Promotion always needs the benchmark gate: at least 3 test documents and 100 reference points on real drawings.

Verifier not yet

Needs	Have	Target
positive labels	2	100
negative labels	0	100
documents with labels	1	10

Detector not yet

Needs	Have	Target
complete pages	0	30
points on complete pages	0	300
documents with complete pages	0	10

Active models

- verifier: none (template matching only)
- detector: none (template matching only)

Dashboard with the readiness cards. Red numbers are below target.

Model	Ready to train when you have	Typical time with 2-3 reviewers
Verifier	100 Correct and 100 Wrong pins, from 10 drawing sets	A few days
Point detector	30 pages marked fully reviewed, 300 pins on them, from 10 drawing sets	About two weeks
Promoting any model	A real test set of at least 3 drawing sets and 100 pins, and beating the template scanner	-

These are guidance: an admin can train earlier, but a model trained on too little data won't pass the promotion gate. **Team** shows who uploaded, scanned and reviewed what, with a per-day table; click a person to see their recent actions.

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Team progress

What each person has uploaded, scanned and reviewed. Totals are all time; the daily table counts review actions (approve/reject). Only admins see this page. Pick a person to see their recent actions.

Days shown 14

2 members

2 active this week

0 drawings uploaded

2 pins reviewed

0 pages fully reviewed

Person	Role	Last active	Uploads	Scans (pages)	Pins reviewed	Approved / rejected	Misses
assistant@example.com	reviewer invite not used	just now	0	0 (0)	2	2 / 0	
boss@example.com	admin	just now	0	0 (0)	0	0 / 0	
Not signed in (CLI / local)		just now	0	1 (1)	0	0 / 0	

Review actions per day

Team progress: per-person totals and review actions per day.

12. Train each legend symbol (Symbol types)

This is the quickest win and needs no special amounts of data: each legend symbol learns from its own Correct / Wrong reviews from whole-set scans.

- 1 Open **Symbol types**. Each tag shows how many of its pins were approved, rejected and added by hand.
- 2 Press **Train** (or **Train every symbol with reviews**). It takes seconds to a few minutes.
- 3 Pinny checks the result on reviews it didn't train on. Learning switches on only if it keeps every correct match. Use **Switch off** / **Switch on** to control it.
- 4 The next **Scan whole set** throws out matches that look like that symbol's rejected examples.

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Symbol types

Each legend symbol learns from its own reviews. Approve correct matches and reject wrong ones from a whole-set scan, then press Train: the next whole-set scan throws out matches that look like that symbol's rejected ones. A symbol's learning is switched on only after it proves itself on reviews it was not trained on (it must not lose correct matches). Pins added by hand are counted but not learned from yet.

Train every symbol with reviews

Tag	Approved	Rejected	Added by hand	Sheets	Learning
D	10	6	0	2	Not trained yet

Train

Symbol types after training: learning on for symbols that passed the check.

13. Build a dataset

- 1 Open **Datasets** and press **Build dataset from current reviews**.
- 2 Wait for it on **Training runs**. The dataset is a frozen snapshot of all reviews, split by drawing set into train, validation and test, so a model is always tested on drawings it never saw.
- 3 Check the **Test split** column: it must say "enough to promote" (3 documents, 100 points) before any model trained on it can be promoted.

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Datasets

A dataset is a snapshot of the reviewed pins and fully reviewed pages, split by drawing into train, val and test.

Build dataset from current reviews Epochs 30

Dataset	Created	Verifier crops (+/-)	Detector pages (points)	Test split
6e38e4a58234 synthetic	1/1/1970, 12:00:00 AM	train 30/63 · val 11/22 · test 0/0	train 3 (30) · val 1 (11) · test 0 (0)	0 documents, 0 points too small to promote

Train verifier

Train detector

Datasets: one row per snapshot, with its train/val/test counts.

Tip. Build a new dataset whenever a good batch of new reviews has come in. Old datasets stay, so you can always compare.

14. Train the verifier and the point detector

- 1 On **Datasets**, set **Epochs** (30 is a good default) and press **Train verifier** or **Train detector** on the dataset's row.
- 2 Follow it on **Training runs**: progress bar, status and a live log. **Cancel** stops it.
- 3 When it finishes, the new model appears on **Models** as a **candidate**. It is not used for scanning until you promote it.

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Training runs

The server shuts down at 7 PM Eastern every night and starts again at 7 AM. A run still going then is interrupted and has to be started again, so start long runs in the morning.

Job	Kind	Status	Progress	Started by	Created
b25d5ae0	Train verifier: dataset eeeeeeeeeeee, 30 epochs	running	27 %	boss@example.com	10/5/2026, 7:01:15 PM
dc0a96ee	Benchmark: dataset dddddddddddd, verifier-20260902T100000Z-0afb405	done	100 %	boss@example.com	10/5/2026, 7:01:15 PM gate: do not promote
163c477f	Benchmark: dataset dddddddddddd, verifier-20260901T100000Z-d4a1ce59	done	100 %	boss@example.com	10/5/2026, 7:01:15 PM gate: promote

Train verifier: dataset eeeeeeeeeeee, 30 epochs
running · started by boss@example.com

Cancel

Log

```

epoch 1: train_loss=1.0000 val_loss=1.2000
epoch 2: train_loss=0.5000 val_loss=0.6000
epoch 3: train_loss=0.3333 val_loss=0.4000
epoch 4: train_loss=0.2500 val_loss=0.3000
epoch 5: train_loss=0.2000 val_loss=0.2400
epoch 6: train_loss=0.1667 val_loss=0.2000
epoch 7: train_loss=0.1429 val_loss=0.1714
epoch 8: train_loss=0.1250 val_loss=0.1500

```

Training runs: a running job with its progress and log.

Important. Training runs on the server's processor and can take hours for the detector. Start long runs in the morning: the server stops at 7 PM Eastern, and an interrupted run must be started again by hand.

15. Benchmark and promote

- 1 On **Models**, press **Benchmark** on the candidate. It runs the candidate and the plain template scanner on the dataset's test drawings.
- 2 Read the side-by-side table: **Precision** (how many of its pins were right), **Recall** (how many real devices it found), and the pass/fail checks under it.
- 3 If the gate says **recommends promotion**, press **Promote**. Promote stays greyed out otherwise.
- 4 To undo, press **Deactivate** on the active model: scans go back to template matching only.

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Models

Active verifier
None

Active detector
None

verifier-20260902T100000Z-0afbf405
candidate

verifier · fake-verifier · trained 9/2/2026, 10:00:00 AM on dataset ddddddddddd · threshold 0.500
Benchmark gate: does not recommend promotion · 12 test pages, 2 documents, 122 points

	Template baseline	Candidate verifier-20260902T100000
Precision	84.8 %	66.7 %
Recall	91.8 %	49.2 %
True positives	112	60
False positives	20	30
Missed (false negatives)	10	62
Test dataset	dddddddddd	dddddddddd

- fail: candidate recall >= baseline recall
- fail: >= 3 test documents

Benchmark again
Promote

verifier-20260901T100000Z-d4a1ce59
candidate

verifier · fake-verifier · trained 9/1/2026, 10:00:00 AM on dataset ddddddddddd · threshold 0.500
Benchmark gate: recommends promotion · 12 test pages, 4 documents, 122 points

	Template baseline	Candidate verifier-20260901T100000
Precision	84.8 %	97.5 %
Recall	91.8 %	96.7 %
True positives	112	118
False positives	20	3
Missed (false negatives)	10	4
Test dataset	dddddddddd	dddddddddd

- pass: candidate recall >= baseline recall
- pass: >= 3 test documents

Benchmark again
Promote

Models: one candidate fails the gate (fewer devices found than the template scanner), the other passes and can be promoted.

16. Scan with the trained models

Once promoted, the models appear in the viewer's **Scan mode** list under **3. Scan with a template**:

Scan mode	Needs	Use it when
Template match	A template box	Always available; the baseline
Template + verifier	A template box and an active verifier	Template matching gives too many false hits; the verifier removes them
Point detector	An active detector, no template	You want devices found without drawing a box

Keep reviewing in every mode: new Correct / Wrong / Missed clicks keep improving the next model you train. A good rhythm is: review for a week, build a dataset, train, benchmark, and promote only when the gate agrees.

17. Optional: read sheet info with OCR

In the viewer's **OCR (admin)** panel, choose **Tesseract** and press **Save**. Everyone then gets **Read sheet info** under each page, which reads the sheet number, title and revision from the title block. OCR does not find devices; it only labels sheets. (The server needs Tesseract installed once.)